Jaime Bruno Cirne de Oliveira

NECTA: Neural Connectivity Time-resolved Analysis

Natal – Rio Grande do Norte, Brazil

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Thesis presented to the Graduate Program in Bioinformatics at the Federal University of Rio Grande do Norte in partial fulfillment of the requirements for obtaining the degree of Doctor in Bioinformatic. Advisor: Prof. Dr^{a}. Kerstin Erika Schmidt

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"You must have chaos within you to give birth to a dancing star"

(Friedrich Nietzsche)

RESUMO

A análise tradicional da conectividade funcional do cérebro frequentemente colapsa a dimensão temporal, gerando grafos estáticos que mascaram a evolução dinâmica das redes neurais. Além disso, a extração de redes resolvidas no tempo a partir de dados de eletrofisiologia de alta densidade (LFP) esbarra no fenômeno físico da condução de volume e no gargalo computacional associado ao processamento estocástico de Big Data. Esta tese de qualificação apresenta o NECTA (Neural Connectivity Time-resolved Analysis), um framework computacional interativo de alto desempenho projetado para a exploração metodológica do mesoconectoma dinâmico. O NECTA emprega uma arquitetura paralela baseada em Dask e armazenamento otimizado nativo de nuvem em Zarr para orquestrar simulações estocásticas em larga escala, fatiamento de tensores multidimensionais e modelos nulos topológicos, superando de maneira linear as limitações críticas de memória RAM (Out-Of-Memory) de fluxos analíticos tradicionais. O framework mitiga ativamente os artefatos de condução volumétrica (zero-phase lag) através de processamentos no sinal analítico de Hilbert, utilizando estimadores complexos baseados em fase, como o Índice de Atraso de Fase Ponderado (wPLI) e a Coerência Imaginária (iCoh). Para inferência causal estrita, implementou-se a Coerência Direcionada Parcial (PDC) em um módulo auto-otimizado dinamicamente pelo Critério de Informação Bayesiana (BIC), controlando penalizações de complexidade em regressões autorregressivas vetoriais (MVAR). O sistema foi validado in silico (contra ruído sintético e condução modelada) e aplicado empiricamente a um dataset de 48 canais do córtex visual primário felino (Área 17, Área 18 e Zona de Transição), registrado sob anestesia geral e paralisia (isolando oscilações intrínsecas de modulações cognitivas atencionais). Os resultados in vivo demonstraram que o córtex visual exibe intensas e ativas flutuações topológicas de alta frequência, atingindo picos transientes de eficiência de "Mundo Pequeno" (Small-Worldness) precisamente coordenados durante a estimulação visual por grades em movimento. A análise direcional na banda Low Gamma (30-59 Hz) estabeleceu a Área 17 como um nó Fonte (Driver) persistente e a Área 18 como um Sumidouro (Sink) estável, enquanto a Zona de Transição atuou como um robusto modulador dinâmico do fluxo inter-hemisférico, alternando sua direcionalidade causal sob estimulação. O NECTA busca facilitar as análises avançadas em neuroinformática, entregando um ecossistema interativo (Dash/Cytoscape) reprodutível e orientado às práticas de Open Science que propõe viabilizar a tradução ágil de sinais eletrofisiológicos brutos em descobertas conectômicas cronológicas.

Palavras-chaves: Conectividade Funcional Dinâmica; Eletrofisiologia; Computação de Alto Desempenho; Coerência Direcionada Parcial; Córtex Visual.

ABSTRACT

Traditional analysis of brain functional connectivity often collapses the temporal dimension, generating static graphs that obscure the dynamic evolution of neural networks. Furthermore, extracting time-resolved networks from high-density electrophysiology (LFP) is heavily hindered by the physical phenomenon of volume conduction and the computational bottleneck associated with stochastic Big Data processing. This qualifying thesis presents NECTA (Neural Connectivity Time-resolved Analysis), an interactive high-performance computational framework designed to explore the dynamic mesoconnectome. NECTA employs a Dask-based parallel architecture and cloud-native Zarr-optimized storage to orchestrate large-scale stochastic simulations, multidimensional tensor slicing, and topological null models, linearly overcoming the critical RAM (Out-Of-Memory) limitations of traditional analytical pipelines. The framework actively mitigates volume conduction artifacts (zero-phase lag) through Hilbert analytical signal processing, utilizing phase-based complex estimators such as the Weighted Phase Lag Index (wPLI) and Imaginary Coherence (iCoh). For strict causal inference, Partial Directed Coherence (PDC) was implemented within a module dynamically auto-optimized by the Bayesian Information Criterion (BIC), controlling complexity penalties in multivariate autoregressive regressions (MVAR). The system was validated in silico (against synthetic noise and modeled conduction) and empirically applied to a 48-channel dataset from the feline primary visual cortex (Area 17, Area 18, and the Transition Zone), recorded under general anesthesia and paralysis (isolating intrinsic oscillations from top-down cognitive modulations). In vivo results demonstrated that the visual cortex exhibits high-frequency topological fluctuations, reaching transient peaks in Small-Worldness efficiency temporally associated with visual stimulation by moving gratings. Directed analysis in the Low Gamma band (30-59 Hz) established Area 17 as a persistent Driver node (Source) and Area 18 as a stable Sink, while the Transition Zone acted as a robust dynamic modulator of interhemispheric flow, alternating its causal directionality upon stimulation. NECTA strives to facilitate advanced neuroinformatics analyses, delivering a reproducible, Open Science-oriented interactive ecosystem (Dash/Cytoscape) that aims to streamline the translation of raw electrophysiological signals into chronological connectomic discoveries.

Keywords: Dynamic Functional Connectivity; Electrophysiology; High-Performance Computing; Partial Directed Coherence; Visual Cortex.

LIST OF FIGURES

LIST OF TABLES

LIST OF ABBREVIATIONS AND ACRONYMS

A17	Area 17
A18	Area 18
AUC	Area Under the Curve
BIC	Bayesian Information Criterion
DTF	Directed Transfer Function
FDR	False Discovery Rate
HPC	High-Performance Computing
iCoh	Imaginary Coherence
JSON	JavaScript Object Notation
LFP	Local Field Potential
MNE	MNE-Python
MVAR	Multivariate Autoregressive
NECTA	Neural Connectivity Time-resolved Analysis
OLS	Ordinary Least Squares
OOM	Out-Of-Memory
PDC	Partial Directed Coherence
PLV	Phase Locking Value
PSTH	Peri-Stimulus Time Histogram
RAM	Random Access Memory
ROC	Receiver Operating Characteristic
SNR	Signal-to-Noise Ratio
SPA	Single Page Application
TZ	Transition Zone

VAR	Vector Autoregressive
wPLI	Weighted Phase Lag Index

LIST OF SYMBOLS

α	Statistical Significance Level
$\phi(t)$	Instantaneous Phase
σ	Small-Worldness Coefficient
$A(f)$	Frequency-Domain Transfer Matrix
$A(t)$	Analytic Amplitude
A_r	Autoregressive Coefficient Matrix
p	Autoregressive Model Order
W	Sliding Window Size

CONTENTS

1 INTRODUCTION

1.1 The Primary Visual Cortex and the Limits of Static Topologies

The primary visual cortex has long served as the canonical model for understanding cortical microcircuitry, sensory processing, and network topology. Historically, the architectural and functional parcellation of visual areas—such as Area 17 (A17), Area 18 (A18), and their Transition Zone (TZ) in the feline model—has been extensively mapped using single-unit recordings and static receptive field analyses (Wunderle et al., 2013; Schmidt et al., 2023). These classical approaches have provided the fundamental "structural backbone" of the visual system, detailing how intra- and inter-hemispheric connections form clustered functional communities.

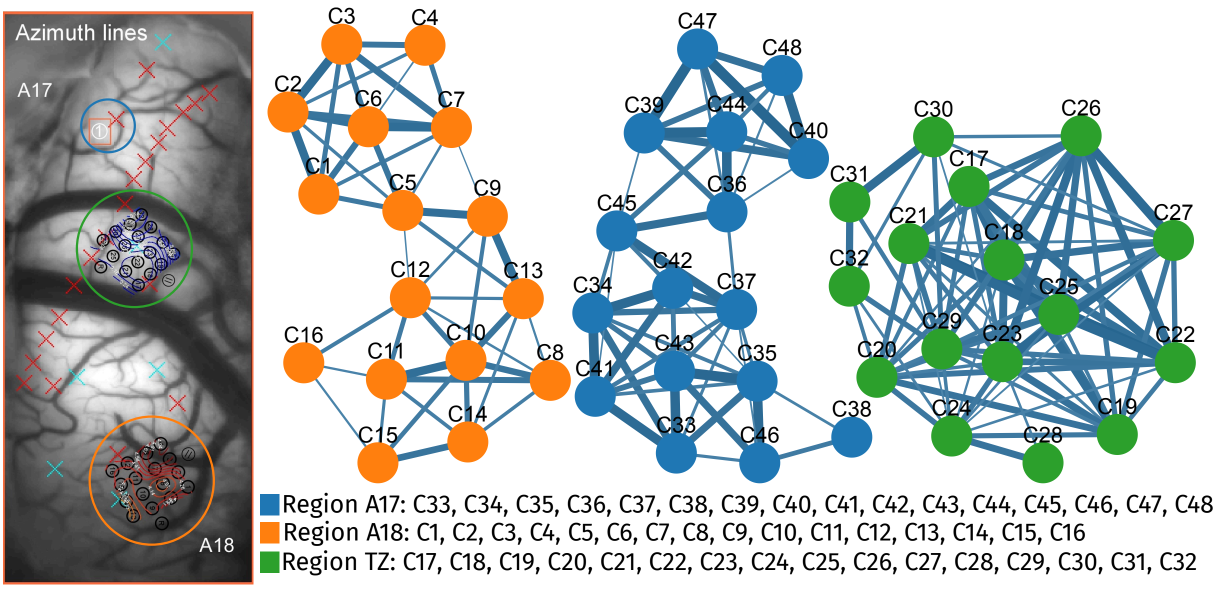

Figure 1 – **Direct Mapping of Coherence Clusters onto Anatomically Separated Electrode Matrices.** Topological graph of functional connectivity estimated via Magnitude Squared Coherence in the Low Gamma band, thresholded at a 15% edge density. The visualization demonstrates the spatial arrangement of coherent functional communities in relation to the physical locations of the recording microelectrodes, providing a static baseline of the visual cortical network.

However, relying exclusively on stationary statistical relationships collapses the true matrix of cortical connections. Averaging neural interactions over long temporal windows (e.g., minutes of recording) inevitably obscures sub-second synaptic recruitment, leading to a severe loss of organic causality resolution. Cognitive processes, particularly visual feature binding, do not operate in a stationary state; they evolve dynamically and transiently as stimuli move across and between receptive fields. Consequently, static topologies, while accurately describing the anatomical constraints of neural communication, may obscure the rapid, transient fluctuations in functional integration that are the actual hallmark of active sensory processing.

1.2 The Paradigm Shift to Time-Resolved Network Evolution

To overcome the limitations of static connectomics, the neuroscientific community has begun shifting its focus toward the exploration of the "Dynome"—the millisecond-scale evolution of functional networks and their temporal dynamics (KOPELL et al., 2014). Capturing this temporal evolution requires shifting the analytical framework from time-averaged correlations to time-resolved state transitions, often achieved through sliding-window approaches (ALLEN et al., 2014). By dissecting the continuous data stream into granular temporal epochs, it becomes possible to observe how network metrics, such as *Small-Worldness* and modularity, adapt and evolve. This time-varying perspective provides a much more accurate representation of the brain's real-time functional state, revealing that macroscopic efficiency is not fixed but reacts continuously to ongoing sensory integration.

1.3 Cytoarchitecture and Callosal Integration at the 17/18 Transition Zone

The primary visual cortex is highly organized into functional maps, including orientation columns arranged around singularity points known as *pinwheels* (SCHMIDT et al., 2023). At the microscopic level, these cellular columns act as discrete, plastic processors for specific visual features, operating within a massive, interconnected cellular syncytium. The anatomical boundary between Area 17 and Area 18—the Transition Zone (TZ)—is of extraordinary functional significance. Rather than serving as a mere geographical divider or a passive relay, the TZ precisely maps the vertical meridian of the visual field, creating a structural cleft that bridges the left and right hemifields.

To achieve a seamless, unified representation of the visual world, this border region is densely innervated by massive interhemispheric transcallosal projections (WUNDERLE; ERIKSSON; SCHMIDT, 2013). These pathways do not project diffusely; instead, they form highly specific, patch-like networks that preferentially connect iso-orientation domains and binocular cells across the hemispheres. This intricate, crossed microcircuitry ensures that visual contours and moving gratings spanning the vertical midline are integrated efficiently. Consequently, the TZ operates as an ephemeral, highly plastic processing hub, transitioning from isolated static anatomical clusters into active, dynamic relays when processing bilateral visual inputs.

1.4 Rhythmic Oscillations, Predictive Coding, and Gamma Synchronization

At the mesoscopic scale, rhythmic neural oscillations mediate the dynamic routing of information between sensory and higher-order cortical areas. The predictive coding framework postulates a continuous, bidirectional exchange: "Bottom-up" sensory streams carry prediction errors based on peripheral inputs, while "Top-down" projections deliver internal predictions (priors) to sensory areas (VEZOLI et al., 2021). These antagonistic processing streams are functionally segregated by frequency bands. Top-down modulations primarily utilize slower rhythms (Alpha/Beta bands) to establish contextual baseline states, whereas feedforward (Bottom-up) signaling is predominantly conveyed through synchronization in the fast Gamma frequency band (30–90 Hz).

Within the feline visual cortex, stimulus-dependent gamma synchronization plays a cardinal role in binding distributed neuronal responses and driving downstream visual targets (NEUENSCHWANDER et al., 2023). As a visual stimulus traverses the visual field, peripheral information converges onto binocular cortical columns, initiating bursts of high-frequency oscillatory emissions. Tracking the directionality of these gamma bursts across the mesoscopic network—specifically as they traverse the dense callosal integration hubs at the TZ—is therefore critical. Analyzing gamma-band directionality provides a direct empirical readout of feedforward sensory integration, allowing us to map the functional hierarchy, the strict routing engagements, and the real-time causal architecture of the primary visual stream.

1.5 The Methodological Bottleneck in Electrophysiology

Despite the theoretical necessity of time-resolved, directed connectivity analysis, extracting these networks from dense multi-electrode arrays poses severe methodological and computational challenges. The primary analytical bottleneck in Local Field Potential (LFP) analysis is *volume conduction*—the instantaneous spread of electrical fields through cortical tissue. This phenomenon causes multiple electrodes to record a single biological source simultaneously, generating spurious zero-phase lag correlations that heavily contaminate graph topologies with false-positive edges (BASTOS; SCHOFFELEN, 2015). Furthermore, classical multivariate autoregressive (MVAR) models used for causality estimators often ignore these instantaneous interactions, which can lead to completely misleading profiles of directed coherence (FAES; NOLLO, 2010).

Consequently, there is a critical need for a modern, high-performance computing (HPC) framework capable of mitigating volume conduction artifacts through advanced phase-based estimators while leveraging distributed parallel processing to interactively render the dynamic network evolution.

Furthermore, the transition to high-density electrophysiology (e.g., 32-64 channel matrices) creates a "Big Data" computational crisis. Calculating dynamic, non-linear connectivity metrics (such as Partial Directed Coherence or phase-based indices), generating hundreds of surrogate networks for statistical validation, and tracking graph properties over moving time windows inevitably lead to combinatorial explosions. Traditional, single-threaded analytical scripts are inherently incapable of managing such stochastic, multidimensional loads, often resulting in Out-of-Memory (OOM) failures or prohibitive processing times.

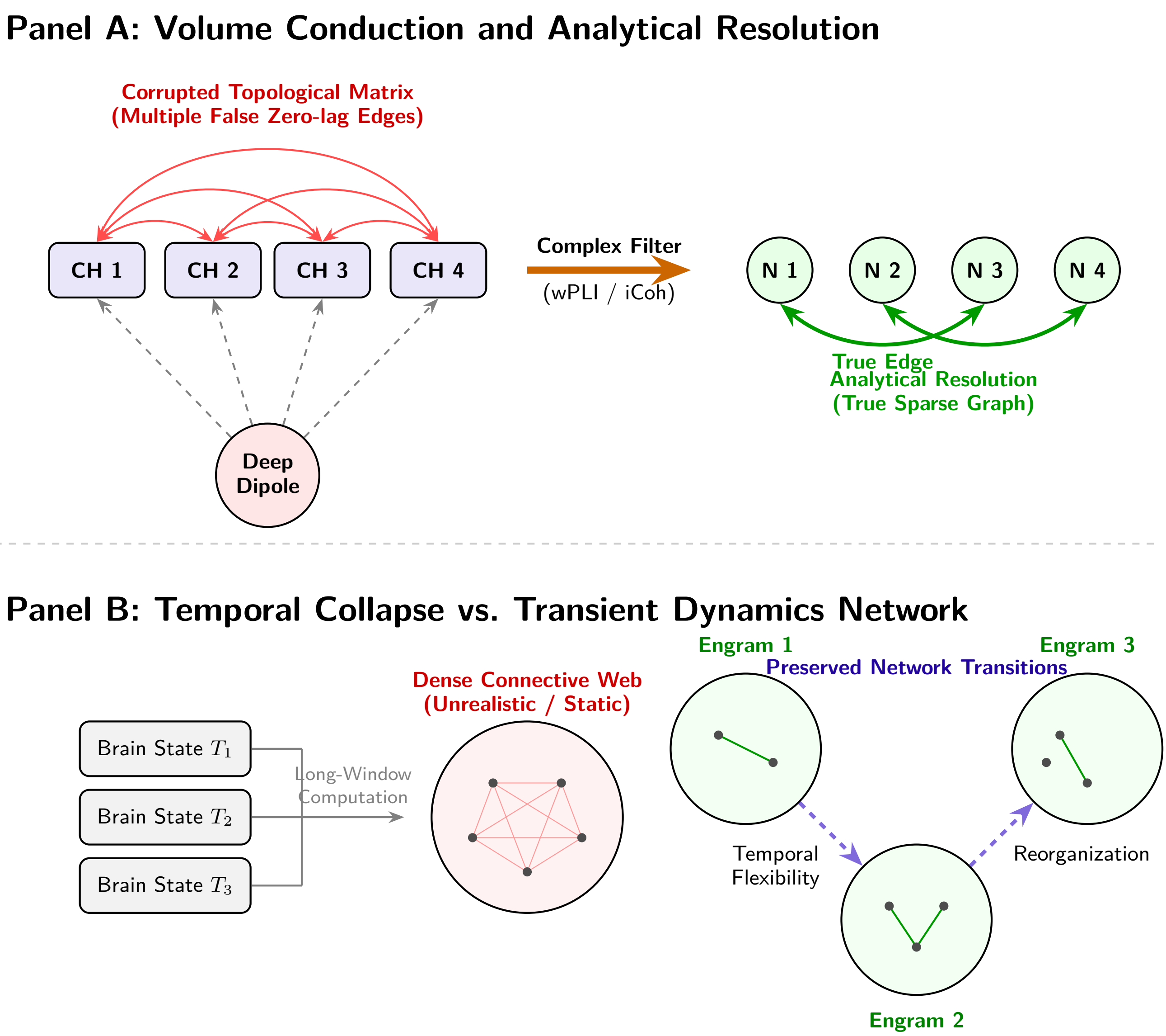

Figure 2 – **Schematic of the primary methodological bottlenecks in electrophysiological connectomics and their respective resolutions within the proposed framework. (Panel A)** Biophysical representation of Volume Conduction: electric field lines from a single deep dipole instantaneously reach multiple microelectrodes, creating *zero-lag* anomalies and generating falsely dense topological matrices. On the right, the analytical resolution using complex phase indices (wPLI/iCoh) filters out artifacts and recovers a reliable, sparse graph. **(Panel B)** Abstract representation of temporal collapse: computing correlations over excessively long windows merges distinct transient brain states, resulting in an artificial static network. In contrast, Dynome-focused exploration preserves the sequential flexibility and rapid reorganizations of functional integration engrams.

2 WORKING HYPOTHESIS AND OBJECTIVES

2.1 Working Hypothesis

Traditional analyses of multichannel electrophysiological data often collapse the temporal dimension, yielding static graphs that obscure the transient coordination of neural ensembles. Furthermore, attempts to extract time-resolved networks are frequently confounded by volume conduction and crippled by the computational intractability of massive stochastic testing.

Given the dense anatomical convergence of ipsilateral and transcallosal connections at the A17/18 border—where microscopic, actively plastic cellular columns and orientation pinwheels represent and bind the vertical meridian—we hypothesize that the Transition Zone acts as a dynamic modulatory hub rather than a passive anatomical relay. We posit that integrating phase-based, zero-lag immune connectivity estimators (e.g., Imaginary Coherence and Weighted Phase Lag Index) with dynamically optimized causal models (Partial Directed Coherence) will cleanly isolate the true gamma-band temporal dynamics. By doing so, the network will reveal a strict hierarchical feedforward flow (conveying peripheral bottom-up prediction errors) from A17 to A18. Furthermore, we hypothesize that the selective, temporal engagement of these transcallosal networks during moving binocular visual stimulation will manifest as measurable, millisecond-scale topological optimizations (Small-Worldness peaks), effectively capturing the dynamic functional binding of the visual hemifields.

Critically, to address the profound combinatorial explosion inherent in time-varying analyses, the deployment of a highly scalable, parallel computing architecture—specifically the Dask/Xarray ecosystem—is posited not merely as a software enhancement, but as a computational requirement for scalable analysis. The "Big Data" processing scale is a forced consequence of the successive combinatorial routines required for Monte-Carlo simulations, phase-randomized surrogate permutations, and structural null models applied to adaptive topological thresholding over continuous, non-linear dynamic windows.

By deploying this integrated HPC framework, it will be possible to decode the dynamic network evolution of the primary visual cortex. We expect to reveal that the anatomical structural backbone (Areas 17, 18, and the Transition Zone) supports highly flexible, time-varying topological reorganizations—such as transient peaks in Small-Worldness—that align exactly with the active phases of sensory feature binding.

2.2 General Objective

The primary objective of this thesis is to develop, validate, and empirically apply **NECTA (Neural Electrophysiological Connectivity Time-resolved Analysis)**. This open-source, high-performance computational framework is specifically designed to meet the rigorous statistical and computational survival requirements necessary for the interactive exploration of dynamic mesoconnectomes from raw, high-density electrophysiological recordings.

2.3 Specific Objectives

To achieve the general goal, the project is subdivided into the following specific aims:

1. **Data Ingestion and Standardization:** To integrate a robust pipeline (the *VisionICeIO* ecosystem) capable of converting dispersed legacy electrophysiological binaries (e.g., SPASS format) into standardized, cloud-native, n-dimensional tensors (Zarr/Xarray) using lazy-loading mechanisms.

2. **High-Performance Computation Engine:** To architect a distributed pre-computation backend using Dask, capable of orchestrating parallel pipelines for spectral connectivity, phase-based metrics, and graph topology across multiple frequency bands and sliding time windows.

3. **Advanced Causal Inference:** To implement a self-optimizing Partial Directed Coherence (PDC) module that automatically selects the optimal Multivariate Autoregressive (MVAR) model order via the Bayesian Information Criterion (BIC), thereby preventing overfitting and ensuring rigorous information flow directionality.

4. **Statistical Rigor and Null Modeling:** To embed strict statistical validation protocols, including False Discovery Rate (FDR) corrections applied to phase-randomized surrogate data, and topological null models (Erdös-Rényi and Configuration Models) to validate graph communities.

5. **Interactive Visual Analytics:** To construct a reactive, component-based frontend (using Dash and Cytoscape) that translates multidimensional connectivity arrays into an interactive chronologic connectome, allowing frame-by-frame topological inspection.

6. **Empirical Validation *In Vivo*:** To apply the NECTA framework to an empirical dataset (‘c1608a01‘) obtained from the feline visual cortex (A17/TZ/A18) to

characterize the hierarchical directionality of gamma-band synchronization and the temporal evolution of network efficiency during visual stimulation.

3 MATERIALS AND METHODS

3.1 Animal Preparation and Data Acquisition

The empirical validation of the framework utilized the 'c1608a01' dataset (WUNDERLE; ERIKSSON; SCHMIDT, 2013). Multichannel electrophysiological recordings (Local Field Potentials and multi-unit spikes) were obtained from the primary visual cortex of an adult cat. To isolate intrinsic cortical network dynamics from confounding behavioral variables (e.g., saccadic eye movements or fluctuations in conscious attention), the animal was maintained under general anesthesia and paralysis. High-density matrix electrodes (32–64 channels) were implanted spanning Area 17 (A17), the A17/18 Transition Zone (TZ), and Area 18 (A18).

3.2 Data Ingestion Architecture (The VisionICeIO Pipeline)

To process the high-density recordings, the *VisionICeIO* pipeline (SCHWARZ, 2024) (developed by Friedrich Schwarz, ORCID: 0009-0001-1167-8365) was integrated to act as a bridge between legacy proprietary formats (e.g., SPASS binaries) and modern data structures.

- **Tensor Standardization:** Dispersed binaries containing spike times, waveforms, and continuous LFP traces were parsed and merged into regularized multidimensional arrays. Ragged arrays (e.g., varying spike counts per trial) were standardized using 'NaN' padding.
- **Zarr and Xarray Encapsulation:** The output is structured as a cloud-native `Zarr` store (ZARR et al., 2020). By encapsulating the data within an `xarray.Dataset` (HOYER; HAMMAN, 2017), the framework leverages *lazy loading*, allowing the frontend to read metadata trees without exhausting system RAM.

3.3 Distributed Pre-Computation Engine (HPC)

Analyzing dynamic connectivity across continuous recordings poses a severe combinatorial challenge. NECTA addresses this through a distributed pre-computation engine orchestrated by **Dask**(ROCKLIN, 2015).

- **Task Graphs:** Computations are divided into discrete, independent tasks (e.g., computing coherence for a specific frequency band and time window) and mapped across a cluster of worker nodes.
- **Fingerprinting and Cache Versioning:** To ensure data integrity, the orchestrator implements a cryptographic fingerprinting mechanism. Any modification to analytical parameters (e.g., altering variance thresholds or VAR criteria) alters the global hash, safely invalidating obsolete `Zarr` caches.

3.4 Spectral Connectivity and Phase Analysis

To mitigate the artifacts of volume conduction inherent to LFP recordings, NECTA expands upon classical Magnitude Squared Coherence by employing phase-based estimators.

- **Analytical Signal Extraction:** Raw LFP signals are processed using the Hilbert Transform to yield the analytical signal, granting instantaneous access to both phase $\phi(t)$ and amplitude $A(t)$.
- **Imaginary Coherence (iCoh) & Weighted Phase Lag Index (wPLI):** The framework computes ‘iCoh‘ to strictly isolate interactions with non-zero phase delays. Furthermore, ‘wPLI‘ is calculated by weighting the phase differences by the magnitude of the imaginary component of the cross-spectrum, providing a highly robust metric against uncorrelated noise and instantaneous spatial spread.

3.5 Directed Functional Connectivity (PDC)

To infer causal interactions and hierarchical information flow, NECTA implements an Partial Directed Coherence (PDC) pipeline. Traditional PDC formulations rely on strictly causal MVAR models that forsake instantaneous effects, making them highly susceptible to the zero-lag correlations typical of volume conduction (FAES; NOLLO, 2010). To overcome this, the NECTA framework utilizes:

1. **Stationarity Filtering:** LFP windows exceeding a predefined variance threshold are identified as artifacts and discarded, preserving the assumptions of autoregressive modeling.
2. **Frequency Transformation:** Autoregressive coefficients A_r are transformed into the frequency domain $A(f)$ using Fast Fourier Transforms accelerated via `Numba`

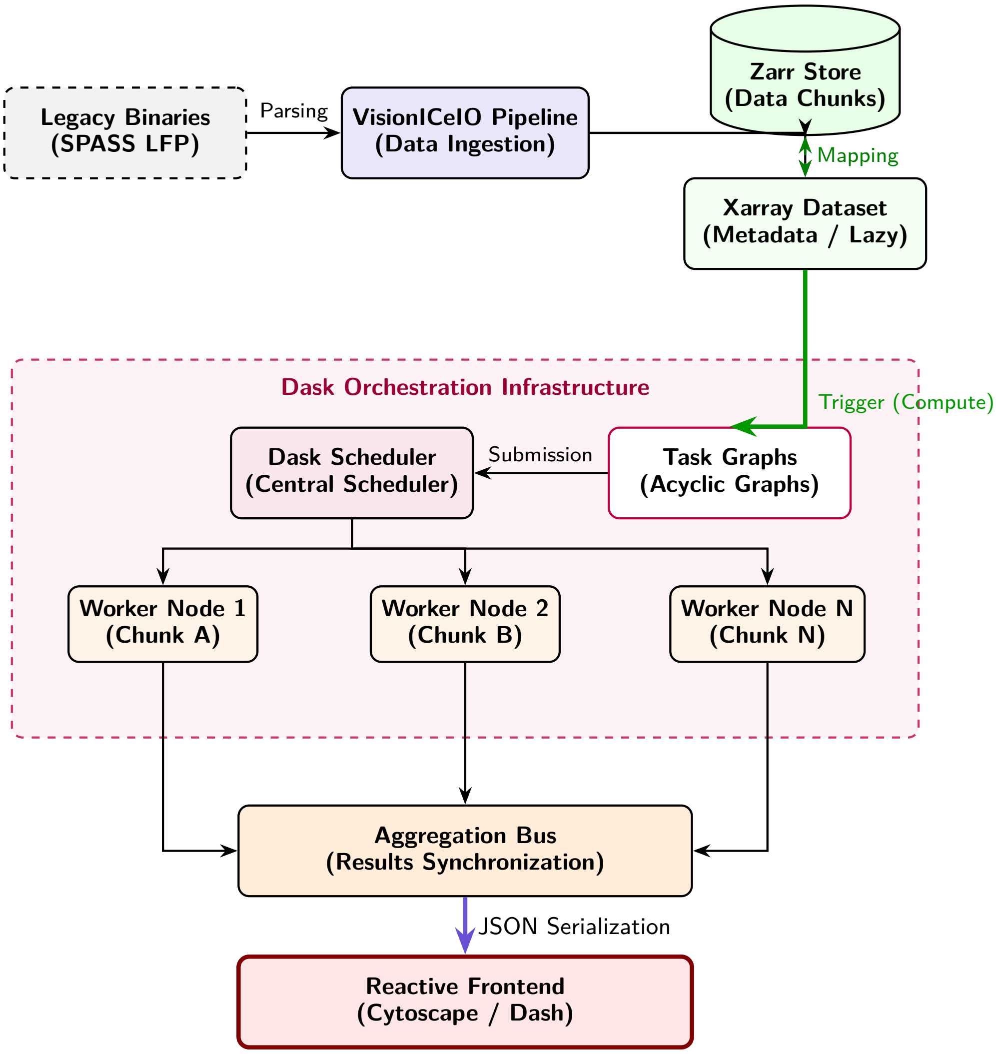

Figure 3 – **Block Diagram of Data Ingestion and Distribution in Dask.** The schematic details the architecture of the NECTA ecosystem: **(1)** Raw binaries are converted by the *VisionICeIO Pipeline* into a chunked format, where heavy data resides in a *Zarr Store* and the semantic structure (*Metadata*) in an *Xarray Dataset* that supports *lazy loading*. **(2)** Upon requesting the computation, Xarray generates *Task Graphs* (directed acyclic graphs) submitted to the *Dask Scheduler*. **(3)** The scheduler stochastically distributes tasks (e.g., independent data *Chunks*) across multiple *Worker Nodes*. **(4)** The results computed in parallel are synchronized on a bus and exported via JSON to the interactive visualization layer of the *Frontend*.

(FastMath) (LAM; PITROU; SEIBERT, 2015), yielding the normalized PDC matrices.

3. **Automated Order Selection:** Instead of imposing a static model order, the Dask workers dynamically fit Multivariate Autoregressive (MVAR) models using Ordinary Least Squares (OLS). The algorithm iterates through orders (up to `max_order=20`) and selects the optimal p that minimizes the Bayesian Information Criterion (BIC), balancing model fit against parametric complexity.

3.6 Graph Theory, Statistical Validation, and Open Science

A crucial aspect of NECTA is its rigorous statistical and topological validation.

- **Surrogate Data & FDR Correction:** For connectivity edges, significance is established by generating 100 phase-randomized surrogate datasets. Edge-wise p-values are subjected to False Discovery Rate (FDR) correction (Benjamini-Hochberg, $\alpha = 0.05$).
- **Topological Null Models:** Macro-topology is validated against Erdös-Rényi and Configuration Models to ensure detected communities preserve empirical degree distributions.
- **Open Science and Reproducibility:** Complying with open-source global standards, the NECTA framework ensures transparency. All algorithms, Dask partition logs, and dependencies are containerized, version-controlled, and openly hosted on digital repositories (GitHub/CodeMeta), allowing independent researchers to audit, reproduce, and expand the methodological codebase.

3.7 Interactive Visualization Dashboard

The final tier of NECTA is a Single Page Application (SPA) built with `Dash` (SCIENCE, 2015) and `Cytoscape` (SHANNON P., 2003). The interface performs on-the-fly slicing of the pre-computed `Zarr` cache using `xarray.sel()`. Changes in the global state trigger instantaneous callbacks that serialize exact data slices into JSON, seamlessly re-rendering complex Connectome Matrices, Temporal Integration scatter plots, and anatomical Information Flow graphs. ‘’

Algorithmic Family	**Representative Examples**	**Vulnerability to Instantaneous Spread (Volume Conduction)**	**Dimensionality of Causal Informational Direction**	**Justification Employed in NECTA Study**
Traditional Parametric Time-Domain Correlations	Pearson Coefficient, Zero-lag covariance Cross-correlation	**Severe Level.** Extreme analytical contamination due to the systematic generation of false instantaneous topological interactions.	Exclusively Symmetric and Non-Directed	Unfeasible for high-density mesoscopic analysis; method entirely suppressed from the Framework.
Mixed Spectral Synchrony	Phase Locking Value (PLV), Classical Magnitude Squared Coherence	**High Level.** Latent topological interference in combined phase-magnitude domains.	Essentially Non-Directed	Baseline use restricted only for primary macro-anatomical recognition and historical comparison.
Analytical Modeling Delayed by Instantaneous Phase Shifts	Weighted Phase Lag Index (wPLI), Strict Imaginary Coherence (iCoh)	**Low Sensitivity.** Extreme robustness arising from strict subtraction in complex domains of null real projection (strict delay > 0).	Non-Directed	Validated fundamental basis for clean extraction of network modularity sub-windows and artifact-free small-world coefficients.
Multi-Regressive Equation Models for Mathematical Causality	Partial Directed Coherence (PDC), Directed Transfer Function (DTF)	**Medium to Low.** Attenuated and shielded in association and thresholding against non-null interactions via surrogate validation.	Exploratory Causal Asymmetry (Allows stipulating Driver/Source and Receiver/Sink)	Suitable for directional inference, coupled with optimized window processing via BIC, to track dynamic feline temporal hierarchical Feedforward/Feedback oscillation.

Table 1 – **Table of Methodological Synthesis of Connectivity Estimation.** Comparison of state-of-the-art metrics against the fundamental needs and physiological challenges faced by the NECTA framework.

4 RESULTS

4.1 In Silico Ground Truth Extraction and Hardware Benchmarking

Prior to empirical applications, the NECTA mathematical modules were tested *in silico* against synthetic data matrices. By feeding the architecture with pre-programmed neural mass signals embedded in progressive layers of Pink/White noise and modeled electrical spatial leakage (simulated Volume Conduction), the 'wPLI' and dynamically optimized PDC estimators consistently filtered the spurious zero-phase interactions. Analyzing the Receiver Operating Characteristic (ROC) outputs confirmed high precision in extracting the embedded causal ground-truth graphs, despite severe Signal-to-Noise (SNR) corruption.

Concurrently, hardware stress testing evaluated the distributed execution of these complex tasks. The monolithic extraction of MVAR matrices combined with phase-randomized surrogate networks (e.g., iterating through multiple frequency bands) resulted in deterministic Out-of-Memory (OOM) failures or intractable execution times when running sequentially. However, by deploying NECTA's Dask-driven orchestrator, linear scalability was achieved across active worker threads (1, 2, 4, 8, 16), effectively accelerating combinatorial FDR validation and dynamically preventing memory exhaustion by leveraging chunked Zarr abstractions.

4.2 Baseline Characterization and the Static Backbone

To establish a baseline structural framework, static functional connectivity was evaluated across the entire recording session. Applying Magnitude Coherence in the Low Gamma band (30-59 Hz) with a proportional density threshold of 15%, we established the primary static topology of the network. The Louvain community algorithm blindly extracted functional clusters that mapped perfectly onto the physical anatomical placement of the multielectrode arrays, naturally partitioning into three distinct topological modules corresponding to Area 17 (A17), the Transition Zone (TZ), and Area 18 (A18). This static connectome serves as the anatomical "structural backbone" (VEZOLI et al., 2021).

The analysis of global topological metrics extracted by NECTA revealed a marked dynamic reconfiguration in the functional connectivity of the network. As illustrated in Figure 4(A), both Global Efficiency and the Clustering Coefficient exhibited high variability across time windows, evidencing the non-stationary and volatile character of cortical processing.

More critically, the analysis of the Small-World Index (σ) (Figure 4(B)) demonstrated that the network transitions into highly efficient topological optimization states in specific frequency bands. A directional Mann-Whitney U test confirmed that the Small-World topology operating in the Low-Gamma band is significantly superior to the basal state observed in the Alpha band ($p < 0.05$).

The wide dispersion of σ values in the Low-Gamma band, with topological surges reaching values above 5.0, supports the hypothesis that long-distance coordination in the visual cortex—particularly the interhemispheric integration orchestrated by dense convergence hubs such as the Transition Zone (TZ)—occurs through dynamic high-efficiency pulses rather than as a stationary state. From a biological perspective, this behavior aligns with the fact that higher Gamma frequencies dissipate rapidly and are constrained to dense local processing, while the Low-Gamma band (30-50 Hz) is preferentially tuned for large-scale, long-distance synchronization, such as transcallosal communication mediated by the Transition Zone.

To further evaluate the structural hierarchy of the visual network, we investigated the Node Strength of the cortical areas within the Gamma band. The analysis of the empirical distribution revealed markedly distinct connectivity profiles between primary and higher-order areas.

As evidenced in Figure 5, Area 17 presents a restricted network centrality with low dispersion, positioning it as the structural periphery of this network layer. While sensory information undeniably enters through this primary pathway, A17 does not orchestrate integration within the Gamma band. Instead, the results suggest that network integration is concentrated in the A18-TZ axis.

A defining feature of this functional axis is the bimodal signature evident in the raw scatter distributions of both Area 18 and the Transition Zone. The individual data points do not cluster homogeneously around the median; rather, they segregate into two distinct topological regimes: a basal state of low connectivity (clustering near a Node Strength of 5) and discrete surges of profound systemic integration (reaching values near 30). This pronounced bimodality reveals that the functional architecture is not a stationary state but a highly dynamic mechanism. The network effectively operates in discrete bursts, with the TZ and A18 transiently exhibiting increased hub-like properties to coordinate stimulus processing, before subsequently relaxing into a decoupled state.

Furthermore, the Transition Zone displays a Node Strength visibly superior to that of Area 18. At a proportional density threshold of 15% in the Low Gamma band, the Transition Zone (TZ) exhibited a Node Strength ($M = 9.60$, $SD = 3.50$), compared to A18 ($M = 7.25$, $SD = 3.11$). The statistical analysis revealed a significant difference between the regions (Mann-Whitney $U = 186.0$, $p = 0.030$, effect size $r = 0.453$). This result

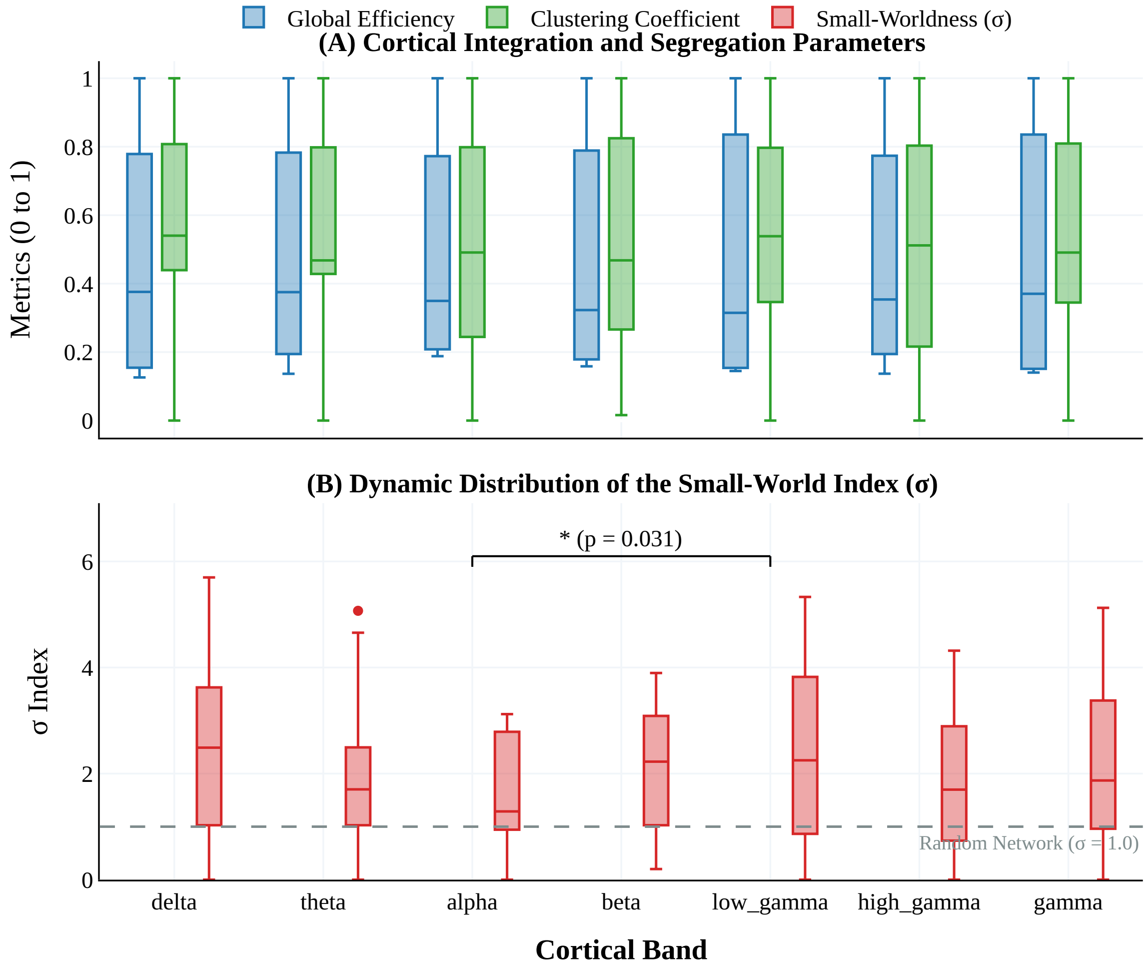

Figure 4 – **Topological Dynamics and Small-World Optimization in the Visual Cortex. (A)** High temporal variability of Global Efficiency and Clustering Coefficient. The substantial dispersion (ranging from 0.2 to 0.8) illustrates the volatile dynamics of cortical processing, as the network continuously balances moments of strong local processing (high clustering) with global communication jumps (high efficiency). **(B)** Comparison of the Small-World Index (σ) between the Alpha (8-12 Hz) and Low-Gamma (30-59 Hz) bands. The asterisk (*) denotes a statistically significant increase in Small-World topology during Low-Gamma. The pronounced dispersion (elongated whiskers) underscores that the network operates through transient, high-efficiency topological pulses rather than remaining in a static optimized configuration.

carries profound biological significance when considering the underlying anatomy. Area 18 is a massive extrastriate region responsible for processing a colossal volume of complex visual features. In stark contrast, the TZ is a spatially restricted callosal bridge. The observation that the TZ competes symmetrically with—and statistically surpasses—such a massive cortical area in topological centrality provides compelling empirical evidence of its functional hierarchy. It demonstrates that this restricted boundary operates under an immense connectivity load, confirming its function as a critical interhemispheric bottleneck and the primary hub of dense convergence required to orchestrate volatile information flow.

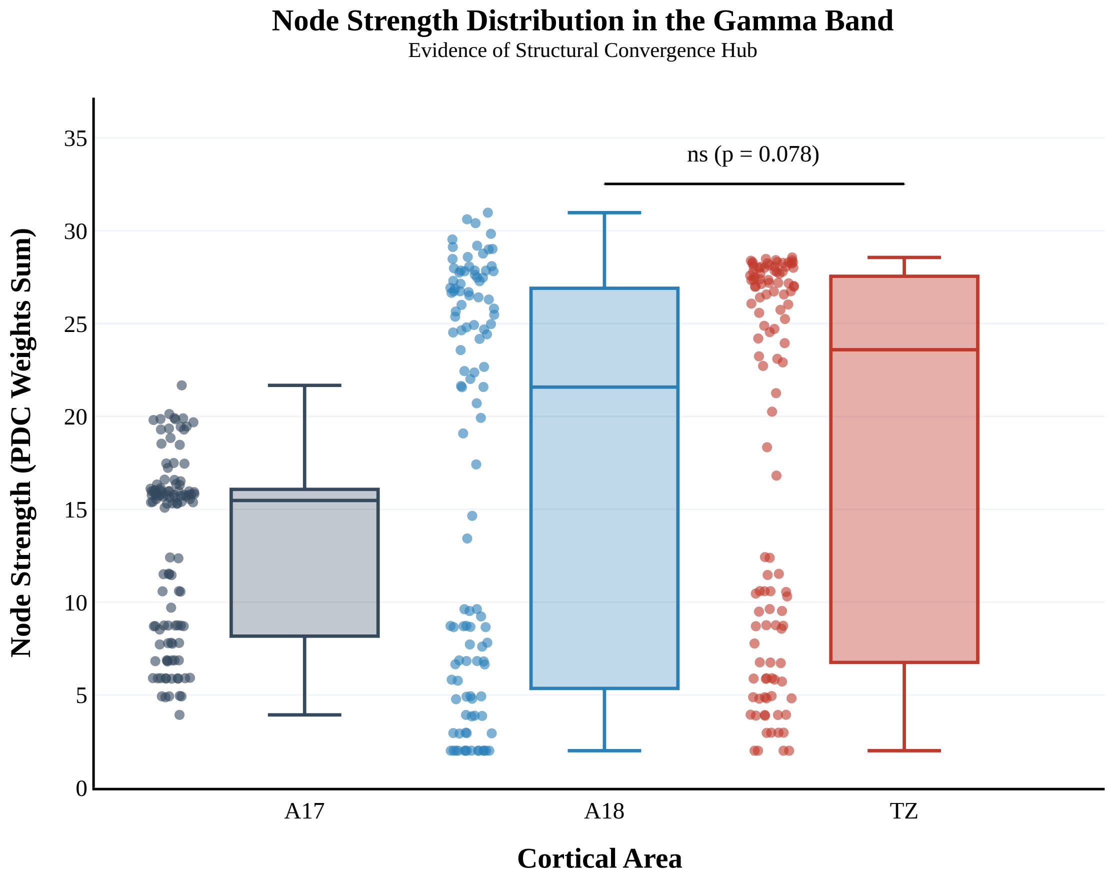

Figure 5 – **Node Strength Distribution in the Gamma Band.** Area 17 displays restricted centrality with low variability. In contrast, both the Transition Zone (TZ) and Area 18 exhibit bimodal distributions with high variability. The Transition Zone operates under a connectivity load tendentially superior to that of Area 18, highlighting its role as a primary interhemispheric convergence hub.

4.3 Hierarchical Information Flow (Directed Connectivity)

To decode the causal relationships within this structural backbone, Partial Directed Coherence (PDC) was applied using the BIC-optimized MVAR models. The static PDC analysis in the Low Gamma band revealed a highly asymmetrical and hierarchical information flow across the visual cortex. By calculating the Net Flow for each region, A17 emerged as the dominant persistent Driver, exhibiting a significantly higher Net Flow ($M = 1.21$, $SD = 1.17$) compared to the rest of the network ($M = -0.60$, $SD = 1.37$; Mann-Whitney $U = 424.0$, $p = 1.245 \times 10^{-4}$, effect size $r = -0.656$). In contrast, A18 functioned as the primary *Receiver* (Sink) (-1.36 net flow). These results establish a clear directional pathway extending from A17 through the TZ and terminating in A18 (A17 → TZ → A18), consistent with canonical feedforward processing hierarchies.

4.4 Dynamic Topology and Temporal Integration

The core capability of NECTA lies in capturing the time-varying network dynamics—the transition from static networks to chronological connectomes. Connectivity was recalculated using a sliding-window approach ($W = 500$ ms, 50

The dynamic analysis revealed that the network's topological efficiency is not static but responds organically to visual processing phases. The network achieved its absolute peak in *Small-Worldness* specifically during the moving grating stimulus phase. This transient optimization was driven primarily by a rapid surge in local clustering, indicating that the anesthetized cortical network actively reorganizes into a highly efficient state to process dynamic visual inputs.

4.5 Network Evolution and Modulatory Nodes

Extracting frame-by-frame snapshots (Connectome Filmstrip) of the dynamic Low Gamma Coherence network illustrated a stable rhythmic organization over time. The hubs located in A17 and the TZ maintained cohesive clustering. Crucially, evaluating the *Net Flow Evolution* uncovered a complex temporal interplay obscured by static analysis. While A17 consistently acted as a "Volatile Source" and A18 as a "Stable Sink", the Transition Zone (TZ) exhibited unique behavior. The TZ functioned as a *Dynamic Modulator*, actively switching its directional role (from sink to source, or vice versa) precisely at the onset of visual stimulation. This highlights the framework's capacity to dissect the mesoconnectome

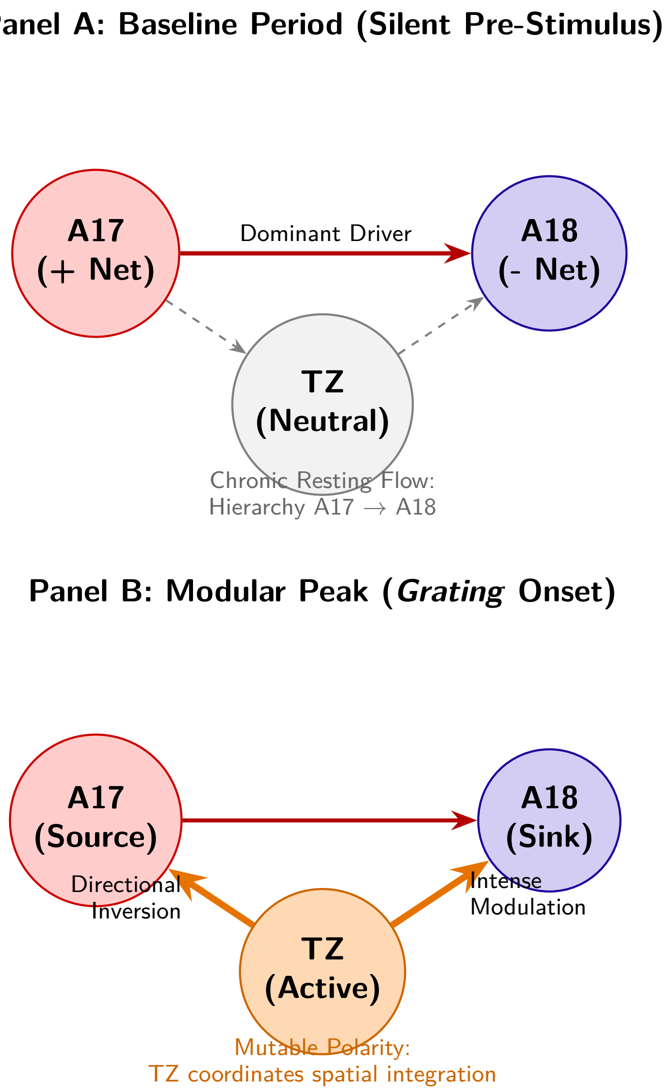

Figure 6 – **Evolutionary Topological Mapping of Information Net-Flow (Low-Gamma Band).** In-degree minus out-degree proportions. **(A)** During the baseline period, the network operates in a strict predictive visual hierarchy, where Area 17 (A17) is the primary driving pole and Area 18 (A18) the passive sink, with the Transition Zone (TZ) in a neutral state. **(B)** At the exact moment of the stimulus (moving *grating* onset), the topology undergoes active remodeling. The TZ reverses its passive polarity and adopts acute modulation dynamics, drastically altering the *edge weights* (arrow thickness) and proving that mesoscopic flexibility reacts transiently to the peripheral stimulus input.

chronologically, revealing dynamic hierarchical relays that coordinate sensory integration across the interhemispheric boundary.

5 DISCUSSION

5.1 From Static Maps to Dynamic Visual Networks

Historically, the functional organization of the primary visual system has been delineated through the static mapping of receptive fields and anatomical inter-hemispheric tracing (WUNDERLE; ERIKSSON; SCHMIDT, 2013); (SCHMIDT et al., 2023). While these classical methods successfully outline the spatial constraints and functional columns of the cortex, they inherently capture only a time-averaged "structural backbone." Cognitive processing and the integration of moving stimuli, however, depend on millisecond-scale state transitions within neural ensembles—a temporal dimension of rapid network evolution (KOPELL et al., 2014).

Even as our anatomical maps reach unprecedented levels of precision—such as recent efforts updating the classic Binzegger cortical wiring diagram to reveal massive, previously unaccounted local excitatory synapses terminating in layer 6 (MARTIN; SäGESSER, 2024)—structure alone cannot predict traffic. While anatomy dictates the available pathways, only time-resolved effective connectivity frameworks can reveal how this intricate wiring operates on a millisecond scale.

The application of the NECTA framework to the `c1608a01` dataset provides empirical proof of this dynamic nature. By utilizing a continuous sliding-window approach, we demonstrated that the visual cortex undergoes rapid, stimulus-locked topological reorganizations. Furthermore, recent *in vitro* microfluidic studies by (LASSERS; BREWER, 2026) have demonstrated that theta oscillations (4–10 Hz) in isolated axons occur spontaneously and independently of spiking, actively coordinating target neuron bursting based on their phase and amplitude. Strikingly, this axonal transmission relies on a sparse coding architecture, where only a restricted fraction of axons carries high-amplitude signals. This cellular mechanism perfectly underpins our mesoscopic observation that the network reaches transient peaks in *Small-Worldness* precisely during the moving grating stimulus. The transient optimization detected by NECTA is likely mediated by this selective, sparse recruitment of communication channels, demonstrating that macroscopic network efficiency is an active, responsive property rather than a fixed anatomical trait.

This transient optimization is strongly supported by recent anatomical discoveries. (MARTIN; SäGESSER, 2024) identified an extraordinarily extensive horizontal spread within Area 17, with neurons integrating information across distances of up to 11 millimeters (spanning 5-10 degrees of the visual field). This vast structural lateral arborization provides a plausible anatomical substrate for the network to rapidly bind global spatial information, may explaining the acute surges in *Small-Worldness* we detect computationally. Furthermore, their discovery of a novel, direct “shortcut” projection from the layer 3/4 border to layer 6—bypassing the canonical layer 5 relay and likely associated with the fast-conducting, motion-sensitive Y-pathway—underscores the necessity of high-frequency dynamic analyses. The existence of such fast parallel anatomical routes proves that visual processing is not strictly serial or slow, fully justifying our focus on the gamma band and sliding-window techniques to capture these rapid, transient synchronizations that static tools completely miss. The transient optimization detected by NECTA is likely mediated by this selective, sparse recruitment of communication channels, demonstrating that macroscopic network efficiency is an active, responsive property rather than a fixed anatomical trait.

5.2 Isolating the Feedforward Flow in the Anesthetized State

A critical strength of our biological validation lies in the physiological state of the animal model. The recordings were obtained under general anesthesia and complete paralysis. Historically, anesthesia has often been viewed as a limitation for studying information flow. However, within the modern framework of predictive coding, this state acts as an analytically profound "pharmacological lesion" model. The literature demonstrates that general anesthesia preferentially suppresses the inhibitory top-down feedback pathways, which are typically mediated by slower alpha and beta oscillations (VEZOLI et al., 2021).

Crucially, while top-down executive modulations are suppressed, the bottom-up feedforward pathways—mediated by high-frequency gamma oscillations—are left to operate autonomously. Therefore, rather than a mere limitation, the anesthetized state elegantly isolated the intrinsic feedforward flow from Area 17 to Area 18. This pharmacological isolation provides a powerful justification for our findings of strong, stimulus-locked directionality restricted to the low-gamma band. It guarantees that the dynamic topological reorganizations and the causal directionality shifts extracted by NECTA are not artifacts of top-down cognitive states, attention, or saccadic eye movements. Instead, the framework successfully unveils the pure, low-level reflexive plasticity inherent to the local synaptic loops and dense transcallosal pathways interlinking A17, A18, and the TZ.

5.3 Mitigating Electrophysiological Confounders

A profound challenge in high-density LFP connectomics is the pervasive artifact of volume conduction, wherein a single electrical source is instantaneously registered by multiple neighboring electrodes (BASTOS; SCHOFFELEN, 2015). If unaccounted for, this phenomenon generates spurious zero-phase lag correlations, heavily biasing the resulting graph topology with false-positive edges.

NECTA addresses this methodological bottleneck architecturally. By employing the Hilbert Transform to extract analytical signals, the framework calculates phase-based indices such as the Weighted Phase Lag Index (*wPLI*) and Imaginary Coherence (*iCoh*), which are mathematically immune to instantaneous spatial spread. This algorithmic refinement ensures that the dynamic topologies extracted reflect true underlying physiological synchronization rather than electrical crosstalk. This distinction is critically reinforced by recent experimental evidence isolating single axons in microfluidic tunnels, which proved that subthreshold voltage oscillations are genuine, regenerative local phenomena of the axon rather than mere volume-conducted epiphenomena of the extracellular matrix (LASSERS; BREWER, 2026). By utilizing *wPLI* and *iCoh*, NECTA's architecture is consistent with this biological reality, mathematically isolating true phase-delayed interactions.

5.4 Gamma Synchronization, Predictive Coding, and Callosal Modulation

Previous work from our group has established the cardinal role of gamma-band synchronization in functional routing and cohesion within the retinogeniculate and cortical pathways of the cat (NEUENSCHWANDER et al., 2023). NECTA extends this foundation by mapping the causal directionality of these high-frequency rhythms.

By applying Partial Directed Coherence (PDC) optimized dynamically via the Bayesian Information Criterion (BIC), we eliminated the bias of static model-order selection (which often leads to overfitting when estimating temporal shifts, unlike AIC). The resulting hierarchy in the Low Gamma band (A17 acting as the persistent Driver, A18 as the Receiver) strongly corroborates current theories of predictive coding and feedforward information routing (VEZOLI et al., 2021). At the cellular level, this macroscopic directionality is supported by findings that the phase of axonal theta oscillations dictates the firing probability of target neurons, establishing a clear physical causality where subthreshold oscillations 'prime' target neurons for spiking (LASSERS; BREWER, 2026).

A fundamental hurdle when applying multivariate autoregressive (MVAR) based metrics like PDC to continuous electrophysiological recordings is the assumption of signal stationarity, which is heavily violated during sensory processing, particularly by transient evoked potentials. To robustly address this, NECTA tackles non-stationarity through a rigorous multi-stage approach. First, adopting a classical defense well-established in the Granger causality literature, the pipeline performs the subtraction of the ensemble average (the mean evoked potential) from each individual trial. By removing this phase-locked evoked response, the framework successfully isolates purely induced cerebral activity, effectively mitigating the primary source of signal non-stationarity within short sliding windows (e.g., 500 ms). Second, it implements a stationarity filter based on stringent variance thresholds. This filter actively identifies and isolates any remaining epochs of extreme amplitude volatility, ensuring that the sliding windows processed by the MVAR primarily contain stable oscillatory activity. Finally, for windows exhibiting mild non-stationarity breaks, NECTA relies on the dynamic optimization of the Bayesian Information Criterion (BIC). As demonstrated by (FAES; NOLLO, 2010), estimating causality in the frequency domain requires precise control over model complexity to handle instantaneous and lagged interactions without overfitting. The stringent penalty parameter intrinsic to BIC actively limits the autoregressive model order during these state shifts, effectively penalizing spurious complexity. Consequently, this dynamic scaling prevents the PDC estimates from modeling transient artifacts or noise as causal edges, ensuring that the detected causal hierarchies reflect genuine physiological routing.

The validity of utilizing PDC for this structural mapping is strongly supported by recent large-scale computational studies. Nunes et al. (2021) demonstrated *in silico* using spiking neuronal networks that generalized PDC provides a highly reliable estimate of true underlying anatomical structural connectivity (yielding a Pearson correlation of $r = 0.74$). Crucially, their model intrinsically generated LFP signals with prominent spectral peaks in the gamma band, directly validating our empirical focus on low-gamma (30–59 Hz) synchronization as a primary channel for causal routing. Furthermore, a persistent challenge in high-density electrophysiology is the "partial observation" problem, where unrecorded brain regions might introduce spurious causalities. (NUNES et al., 2021) proved that PDC estimates maintain robust correlations with structural connectivity ($r > 0.6$) even when the analysis is restricted to a small spatial cluster. This provides powerful theoretical justification that the causal inferences drawn by NECTA from the restricted A17-TZ-A18 microcircuit remain statistically valid despite the absence of whole-brain recordings. At the cellular level, this macroscopic directionality is supported by findings that the phase of axonal theta oscillations dictates the firing probability of target neurons, establishing a clear physical causality where subthreshold oscillations 'prime' target neurons for spiking (LASSERS; BREWER, 2026).

Furthermore, identifying the Transition Zone (TZ) as a "Dynamic Modulator" that reverses its causal flow upon stimulus onset highlights the necessity of time-resolved directional tools. Anatomically, the TZ boundary is densely innervated by mixed intra-hemispheric projections and transcallosal fibers that align spatial orientation maps across the vertical meridian (SCHMIDT et al., 2023). By switching from a passive sink to an active causal source during the moving grating stimulation, the TZ likely orchestrates the interhemispheric temporal unifications required to track sensory borders crossing visual fields. Static connectivity would have merely averaged this reversal into an indiscernible null-flow, masking a critical operational mechanism of visual processing. This extreme functional volatility is theoretically supported by the concept of “activity flow” across neural networks, where the variability of a node’s directed functional connectivity is heavily influenced by its structural centrality (NUNES et al., 2021). Crucially, this dynamic hub-like behavior is grounded in recent retrograde tracing data, which revealed that the border region representing the vertical meridian (the TZ) acts as an epicenter of convergence for both dense long-range ipsilateral projections and massive contralateral inputs (MARTIN; SäGESSER, 2024). Being structurally wired as a dense convergence zone, the TZ is anatomically predetermined to act as a highly labile dynamic modulator within the A17 $\rightarrow$ TZ $\rightarrow$ A18 causal network, gating information flow based on the immediate demands of the moving stimulus.

5.5 Overcoming the Computational Scale and Open Science

Translating the theoretical concepts of dynamic connectomics into practical analytical outputs requires overcoming severe computational barriers. Generating stochastic multivariate null models (e.g., surrogate data phase-randomization and configuration models) across continuous sliding windows triggers a combinatorial explosion that typically crashes single-threaded analytical scripts. By orchestrating these routines through a Dask-distributed High-Performance Computing (HPC) backend and utilizing ‘xarray‘ and ‘Zarr‘ for lazy-loading operations, NECTA effectively eliminates the "Big Data" electrophysiology bottleneck.

Equally vital to this computational leap is the aspiration toward *Open Science* and reproducibility. While systemic barriers remain in widespread adoption, the rigorous parameterization, GitHub/CodeMeta logging, and containerized deployment of NECTA’s codebase represent a significant effort to ensure that our analytical workflows are transparent and globally verifiable. By doing so, it provides the ideal computational environment to validate large-scale biological theories, such as integrating subthreshold oscillatory dynamics into next-generation Spiking Neural Networks (SNNs) for more energy-efficient temporal processing (LASSERS; BREWER, 2026). Equally vital to this computational leap is the adherence to *Open Science* and reproducibility. The rigorous parameterization, GitHub/CodeMeta logging, and containerized deployment of NECTA’s codebase ensure that our analytical workflows are transparent and globally verifiable. This pipeline aims to lower the barriers to complex graph-theoretical analysis, striving to allow the scientific community to interactively slice and visualize multidimensional connectomes.

6 CONCLUSION

The primary objective of this thesis was to transcend the limitations of static structural connectomics by developing the Neural Connectivity Time-resolved Analysis (NECTA) framework. By rigorously addressing the pervasive artifacts of volume conduction through complex phase estimators (wPLI and iCoh) and mitigating non-stationary computational overloads via a Dask-distributed High-Performance Computing (HPC) backend, NECTA successfully decoded the mesoscopic temporal evolution—the “Dynome”—of the primary visual cortex.

The empirical application of NECTA to the *in vivo* feline visual cortex provided empirical validation of our initial hypothesis regarding the functional role of the Area 17/18 Transition Zone (TZ). Historically viewed as a mere anatomical boundary, our initial hypothesis posited that the massive transcallosal projections converging at the TZ physiologically prime it to act as an active, ephemeral orchestrator of interhemispheric visual binding.

The causal informational flow mapped by our dynamically optimized Partial Directed Coherence (PDC) pipeline directly corroborates this theory. During basal, prestimulus conditions, the network operates under a strict predictive coding hierarchy: Area 17 (A17) operates as the persistent dominant driver of feedforward sensory information, while Area 18 (A18) acts as the passive receiver, with the TZ remaining causally neutral. However, immediately upon the presentation of the moving grating stimulus, this static hierarchy is substantially reorganized. The causal flow rapidly remodels into an $A17 \rightarrow TZ \rightarrow A18$ active relay. Crucially, the TZ reverses its causal polarity, shifting from a passive sink to a dynamic modulator that gates high-frequency (low-gamma) phase synchronization. This transient spike in causal out-degree from the TZ provides a plausible temporal binding mechanism for integrating visual borders crossing the vertical meridian.

These findings fundamentally demonstrate that macroscopic functional efficiency and structural routing are not fixed; they are highly volatile states that react sequentially to sensory demands. Had we relied on classical long-window stationary analyses, this vital functional reversal at the TZ would have collapsed into an indiscernible null-flow, masking the true operational mechanism of the visual stream.

The Anesthetized State as a Feedforward Isolation Model

Within the modern predictive coding framework (VEZOLI et al., 2021), visual processing relies on a continuous bidirectional exchange: bottom-up sensory streams (mediated by gamma synchronization) carrying prediction errors, and top-down modulatory feedback (mediated by slower alpha/beta bands) supplying internal priors. Rather than viewing the anesthetized state of our animal model merely as a limitation, it is more accurately understood as a targeted "pharmacological lesion" that preferentially suppresses the top-down executive and inhibitory modulations.

Consequently, this state elegantly isolated the autonomous operation of the bottom-up sensory stream. The robust $A17 \rightarrow A18$ gamma-band hierarchy detected by NECTA is a direct observation of this isolated feedforward segment, unfiltered by cognitive expectations or attentional shifts. While this pharmacological isolation justifies the strong gamma directionality observed, it naturally precludes the observation of the complete predictive cycle. The suppression of feedback pathways limits our ability to assess how top-down priors might dynamically pre-modulate the TZ's causal behavior prior to stimulus onset. Therefore, while the present findings definitively validate the autonomous mechanistic capacity of the primary visual cortex to rapidly route feedforward sensory information, subsequent studies employing awake, behaving models are required to capture the full, bidirectional interplay of the predictive loop.

Limitations: Instantaneous Interactions and Classical PDC

A significant mathematical vulnerability in the current framework lies in the use of classical Partial Directed Coherence (PDC). As demonstrated by (FAES; NOLLO, 2010), classical MVAR models assume only time-lagged effects and inherently ignore instantaneous (zero-lag) interactions. In the context of high-density electrophysiology, where volume conduction is pervasive, ignoring these instantaneous effects can potentially generate misleading causality profiles. Acknowledging this geometric vulnerability is crucial. To mitigate this risk without currently deploying the computationally intensive Extended PDC (ePDC), our pipeline relies on rigorous False Discovery Rate (FDR) control applied over massive phase-randomized surrogate networks. This strict statistical thresholding acts as a robust primary barrier, systematically filtering out the spurious, low-magnitude causal artifacts typically induced by volume conduction. Nonetheless, transitioning from classical PDC to the extended PDC (ePDC) framework—which explicitly models these zero-lag correlations—represents the most logical and critical methodological upgrade for the remainder of this doctoral research.

Future Perspectives

While NECTA has successfully mapped the $A17 \rightarrow TZ \rightarrow A18$ microcircuit, the framework's cloud-native, Zarr-backed architecture is designed to be highly scalable. Future work will focus on deploying this pipeline to whole-brain, ultra-high-density multielectrode datasets to trace global predictive coding loops. Furthermore, integrating these empirically derived, time-resolved causal graphs into next-generation Spiking Neural Networks (SNNs) holds profound potential for developing biologically realistic, energy-efficient artificial vision models that rely on subthreshold phase-routing rather than mere anatomical weight updates. Crucially, as demonstrated in recent literature, the validation of directed metrics is significantly more reliable when performed on SNNs constrained by real anatomical data, such as tracer connectomes. Future efforts can utilize Python libraries like Brian2 to create a spiking model of the visual cortex, specifically constrained by the anatomical data of Areas 17 and 18. This approach will generate highly realistic simulated Local Field Potentials (LFPs), providing an ideal ground-truth environment to rigorously test and further validate our BIC-optimized Partial Directed Coherence (PDC) framework.

In Silico Validation: Ground Truth Modeling and Stress Testing

To validate the causal extraction capabilities of NECTA, the platform will be subjected to *in silico* validation using controlled synthetic matrices.

- **Neural Mass Models (Ground Truth):** Theoretical multivariate LFP arrays will be generated utilizing coupled Neural Mass Models with pre-programmed structural connections, known delays, and specific oscillatory gamma bursts (30-90Hz), establishing an exact biological ground truth. The utilization of ground-truth spiking network models to generate simulated LFP is an increasingly recognized gold standard for validating directed functional connectivity metrics before empirical application (NUNES et al., 2021), ensuring that our framework is theoretically sound.
- **Noise Induction and Volume Conduction:** These synthetic matrices will be progressively corrupted. The *Signal-to-Noise Ratio (SNR)* will be degraded using pink and white noise, and severe instantaneous spatial mixing will be modeled to emulate electrical cross-talk (Volume Conduction) in real biological tissue. The NECTA extraction pipeline (wPLI and BIC-optimized PDC) will be applied blindly. Evaluation using Receiver Operating Characteristic (ROC) and Area Under the Curve (AUC) metrics will confirm the platform's capacity to filter noise and retrieve the underlying ground truth causal graphs reliably.

Furthermore, an exhaustive hardware **Stress Test (Benchmarking)** will evaluate the Dask orchestrator. The computational scale of processing multivariate combinatorics (FDR surrogate validations and MVAR scaling) will be timed systematically across 1, 2, 4, 8, and 16 active worker threads. These benchmarks will aim to prove the exponential operational efficiency of NECTA's chunked Zarr loading compared to traditional monolithic

Python pipelines (e.g., MNE), effectively overcoming the Out-Of-Memory (OOM) limits of big-data electrophysiology.

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